EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics

Fall 2025

EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

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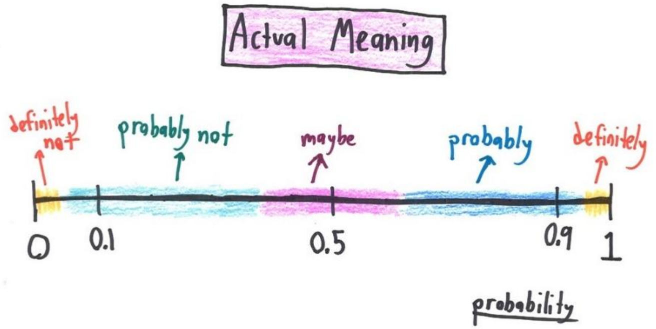
Week 9 - Lecture No 17 – October 15, 2025



Announcements!

1. All lectures posted.
2. Scores for all 4 quiz and MT posted.

Agenda for today

1. Quiz No 05
2. Midterm Exam
3. Unit 4 - Discrete Random Variables

Midterm Exam ?

ID	Name	CLO-1 Q1 (13 Pts)	CLO-5 Q2 (10 Pts)	CLO 2 Q3 (12 Pts)	CLO 4 Q4 (15 Pts)	**Total (50 Pts)**
	Averages	10.82	8.61	6.89	12.93	**39.25**
	Percentage	83.24	86.07	57.44	86.19	**78.50**

Unit 4
Discrete Random Variables

Multiple random variables and joint PMFs

marginal pmfs

- $X : p_X$
- $Y : p_Y$

$\mathbf{P}\,(\,X = \,Y\,) = 2/20$

Joint PMF: $p_{X,Y}(x, y) = \mathbf{P}\,(\,X = x \text{ and } Y = y\,)$

$$\sum_x \sum_y p_{X,Y}(x, y) = 1$$

Y = Second roll				
4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = First roll $p_X(3)$

$p_{X,Y}(1, 3) = 2/20$

$p_X(4) = 1/20 + 2/20$

$p_Y(2) = (1 + 3 + 1)/20$

$$p_X(x) = \sum_y p_{X,Y}(x, y)$$

$$p_Y(y) = \sum_x p_{X,Y}(x, y)$$

Functions of multiple random variables

$$Z = g(X, Y)$$

$$\Rightarrow p_Z(z) = \mathbf{P}(Z = z) = \mathbf{P}(g(X, Y) = z) = \sum_{(x,y):g(x,y)=z} p_{X,Y}(x, y)$$

Expected value rule:

$$\Rightarrow E[g(X, Y)] = \sum_x \sum_y g(x, y)\, p_{X,Y}(x, y)$$

Linearity of expectations

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y]$$

$\Rightarrow$ $E[X+Y] = E[g(x, y)]$

$\Rightarrow$ $g(x, y) = x + y$

$\Rightarrow$ $= \sum_x \sum_y (x+y)\, p_{X,Y}(x,y)$

$\Rightarrow$ $= \sum_x \sum_y (x)\, p_{X,Y}(x,y) + \sum_x \sum_y (y)\, p_{X,Y}(x,y)$

$\Rightarrow$ $= \sum_x x \sum_y p_{X,Y}(x,y) + \sum_y y \sum_x p_{X,Y}(x,y)$

$\Rightarrow$ $= \sum_x x\, p_X(x) + \sum_y y\, p_Y(y) = E[X] + E[Y]$

Linearity of expectations

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y]$$

$$\boldsymbol{E[X_1 + \ldots + X_n] = E[X_1] + \ldots + E[X_n]}$$

$$\mathbf{E}[\,2X + 3Y - Z\,] = E[2X] + E[3Y] - E[Z] = 2E[X] + 3E[Y] - E[Z]$$

The mean of the binomial

$$E\,[\,X\,] = \sum_{k=0}^{n} k p_X(k) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}$$

- X: binomial with parameters n, p
  - ❖ number of successes in n independent trials

$X_i = 1$ if i^{th} trial is a success; p
$X_i = 0$ otherwise $1-p$
(indicator variable)

$$E\,[\,X\,] = np$$

$$X = X_1 + \ldots + X_n$$

$$E[\,X\,] = \underbrace{E[X_1]}_{p} + \ldots + \underbrace{E[X_n]}_{p} = np$$

Exercise No 02!

Example (n = 5)

$\{H_1, T_2, T_3, T_4, H_5\}$

X =
Y =
Z =

Suppose you toss a biased coin n times with parameter 'p' (for Head). Define the following random variables X, Y and Z:

X = The number of heads in n throws

$Y = 3X + 4$

$Z = X^2$

- Calculate Expected value of X, Y and Z.

- $E[X] = np$

- $E[Y] = 3np + 4$

$$E[Z] = \sum_{k=0}^{n} k^2 \binom{n}{k} p^k (1-p)^{n-k}$$

Conditional PMFs

$A = \{ Y = y \}$

$$p_{X|A}(x) = \mathbf{P}(X = x \mid A)$$

$$p_{X|Y}(x \mid y) = \mathbf{P}(X = x \mid Y = y) = \frac{p\,(X = x,\, Y = y)}{p\,(Y = y)}$$

$$p_{X|Y}(x \mid y) = \frac{p_{X,Y}(x,y)}{p_Y(y)}$$ defined for y such that $p_Y(y) > 0$

$$\sum_x p_{X|Y}(x|y) = 1$$

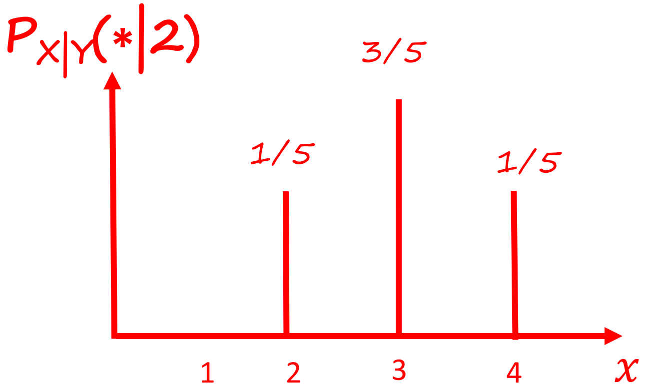
Y = Second roll; X = First roll

Y \ X	1	2	3	4
4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		

$P_Y(2) = 5/20$

$P_{X|Y}(1|2) = 0$

$P_{X|Y}(2|2) =$



$$p_{X,Y}(x,y) = p_Y(y)\,p_{X|Y}(x \mid y)$$

$$p_{X,Y}(x,y) = p_X(x)\,p_{Y|X}(y \mid x)$$

Conditional PMFs involving more than two r.v.'s

- **Self-explanatory notation**

$$\Rightarrow \quad p_{X|Y,Z}(x|y,z) = p\,(X = x \mid Y = y,\, Z = z) = \frac{p\,(X = x,\, Y = y,\, Z = z)}{p\,(Y = y,\, Z = z)}$$

$$\Rightarrow \quad = \frac{p_{X,Y,Z}(x,y,z)}{p_{Y,Z}(y,z)}$$

$$\Rightarrow \quad p_{X,Y|Z}(x,y|z) = p\,(X = x,\, Y = y \mid Z = z)$$

- **Multiplication rule**

$$\mathbf{P}\,(A \cap B \cap C) = \mathrm{P}\,(A)\,\mathrm{P}\,(B \mid A)\,\mathrm{P}(C \mid A \cap B)$$

$A = \{X = x\}$ $B = \{Y = y\}$ $C = \{Z = z\}$

$$p_{X,Y,Z}(x,y,z) = p_X(x)\; p_{Y\,|\,X}(y \mid x)\, p_{Z\,|\,X,Y}(z \mid x,y)$$

Conditional Expectation

$A = \{ Y = y \}$

$$\mathbf{E}[\, X \,] = \sum_x x \, p_X(x) \qquad \mathbf{E}[\, X \mid A \,] = \sum_x x \, p_{X|A}(x) \qquad \mathbf{E}[\, X \mid Y = y \,] = \sum_x x \, p_{X|Y}(x \mid y)$$

Expected Value Rule

$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x) \qquad \mathbf{E}[g(X) \mid A] = \sum_x g(x) p_{X|A}(x)$$

$$\mathbf{E}[\, g(X) \mid Y = y \,] = \sum_x g(x) \, p_{X|Y}(x \mid y)$$